

When an 8-K meets unusual option skew

Gator Quant Hacks 2026 · Systematic Trading · Massive

1. Hypothesis

For a non-specialist: an 8-K tells investors that a company has publicly disclosed something. We ask whether option prices just before that disclosure contain a useful clue, and whether the filing helps separate those moments from ordinary days. The filing is context, not a forecast of whether the share price will rise or fall.

Item 7.01 is the Regulation FD section of Form 8-K. Companies may use it to make information broadly public after a selective communication. Its varied subjects give event timing, not a fixed bullish or bearish meaning. [Investor.gov: How to Read an 8-K](#).

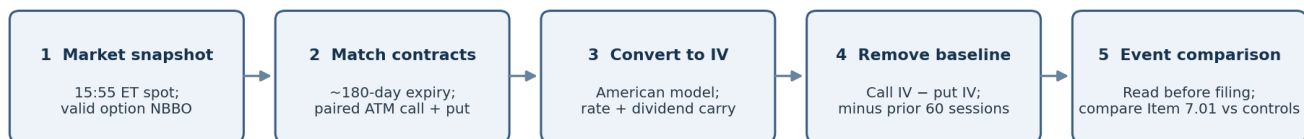
Portfolio-manager hypothesis. A pre-filing relative-skew residual may capture option-market repricing around information about to become public. Conditioning on an actual disclosure may separate those episodes from ordinary option noise; a same-company placebo tests whether the filing adds information beyond skew alone. The null is a zero net event-minus-placebo return after costs. This is a price proxy, not position data; historical open interest is unavailable and volume is not substituted for it.

2. Method: build IV180

For each company-session, spot is the last completed five-minute bar at 15:55 ET. The pipeline selects a standard expiry nearest 180 calendar days (within 150–210), then the strike nearest spot with both a call and a put. Each option uses a valid bid/ask midpoint no more than 60 seconds old from the preceding five minutes. QuantLib solves American implied volatility using the prior Treasury yield and trailing paid cash dividends as carry. “IV180” is a maturity target, not an exact 180-day contract.

Implied volatility is the model input that matches an observed option price. The signal is matched call IV minus put IV, less the company’s trailing 60-session average using earlier sessions only, then standardized across companies. A high-skew z-score above 0.43 is a nominal upper-third cutoff, not an exact empirical tercile. Daily option bars, rather than IV snapshots, mark trade outcomes.

From option quotes to a pre-filing signal



IV180 is a maturity target, not an exact 180-day contract. The IV quotes form the signal; daily option bars mark trade outcomes.

Figure 1. Matched option quotes build the predictor; daily bars later mark modeled trade outcomes.

3. Trade and results

The event is tradeable only from the first session after filing, avoiding assumptions about intraday timing. The primary trade holds 100 shares and buys one 3–6 month put about 5% below spot, exiting after 10 sessions. It limits downside while keeping upside exposure. Five standard structures are tested: long call, covered call, protective put, collar, and cash-secured put. Four bonus structures are risk reversal, bull call spread, short strangle, and short straddle. Costs are 2% of option premium per leg per side and 3 basis points per stock side.

The in-sample filing/signal window is March 7, 2022–December 31, 2024. Saved high-skew Item 7.01 protective-put means are 0.72% of entry spot at 10 sessions and 1.15% at 21 sessions; matched high-skew placebo means are 0.37% and 0.75%. The gaps are 0.35 and 0.40 percentage points, with t-statistics 0.62 and 0.44: positive, but too uncertain to establish a reliable improvement. These saved values predate the new requested-end-date guard and need a bounded refresh before final performance claims.

Portfolio measure	Saved in-sample result
Annualized Sharpe (33 monthly cohorts)	0.45
Summed cohort P&L / maximum drawdown	+11.6% / 6.54%
Positive cohorts	42%

At 21 sessions, risk reversal averages 0.81% of entry spot, above four of five standard structures, but below protective put at 1.15%. Trade-level P&L; is scaled by entry spot; Sharpe is a separate monthly-cohort statistic. The protective-put event/placebo gap changes sign by year.

Saved 2022–2024 development results · 2% option-premium haircut

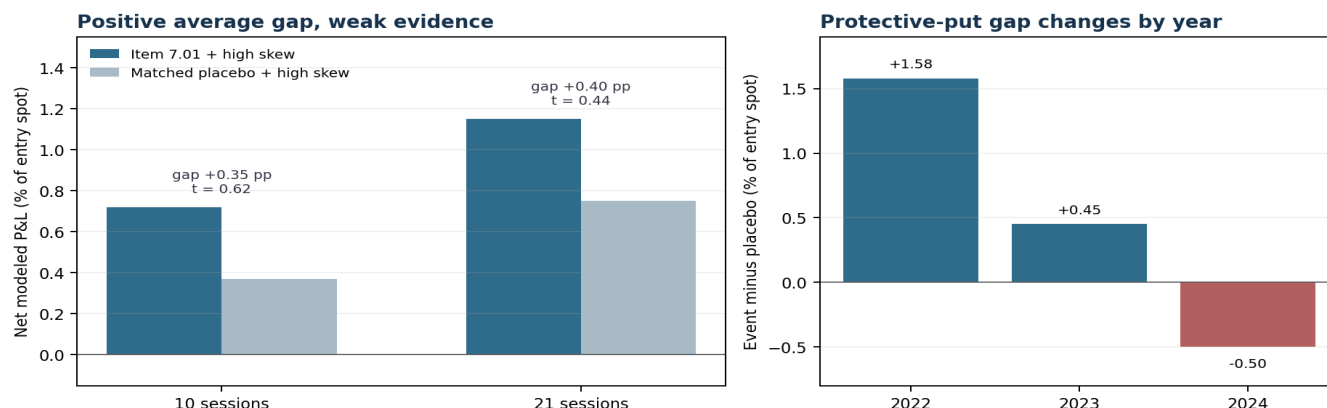


Figure 2. Pooled event/control results and year-by-year protective-put gap. The 2024 gap is negative; the pooled effect is not persistent.

4. Risks and failure conditions

The sample is small; the static large-cap universe may have survivorship bias. Daily option bars are imperfect fill proxies, and the cost model may miss spreads and market impact. The put can lose premium in quiet markets. The saved annualized Sharpe is 0.45, below the requested target; consistent profitability is not established. Treat this as exploratory.

5. Trade specification and reproduction

Use the high-skew Item 7.01 group, enter the first session after filing, buy 100 shares plus one 3–6 month put about 5% below spot, and exit after 10 sessions. Run **strategy.ipynb** with the pinned requirements and **MASSIVE_API_KEY**. Expiry, entry-timing, and filing-category sensitivity checks remain pending. The evaluation configuration is Jan 2, 2025–Oct 2, 2026, with a 100-calendar-day IV warm-up; it remains disabled and was not run.